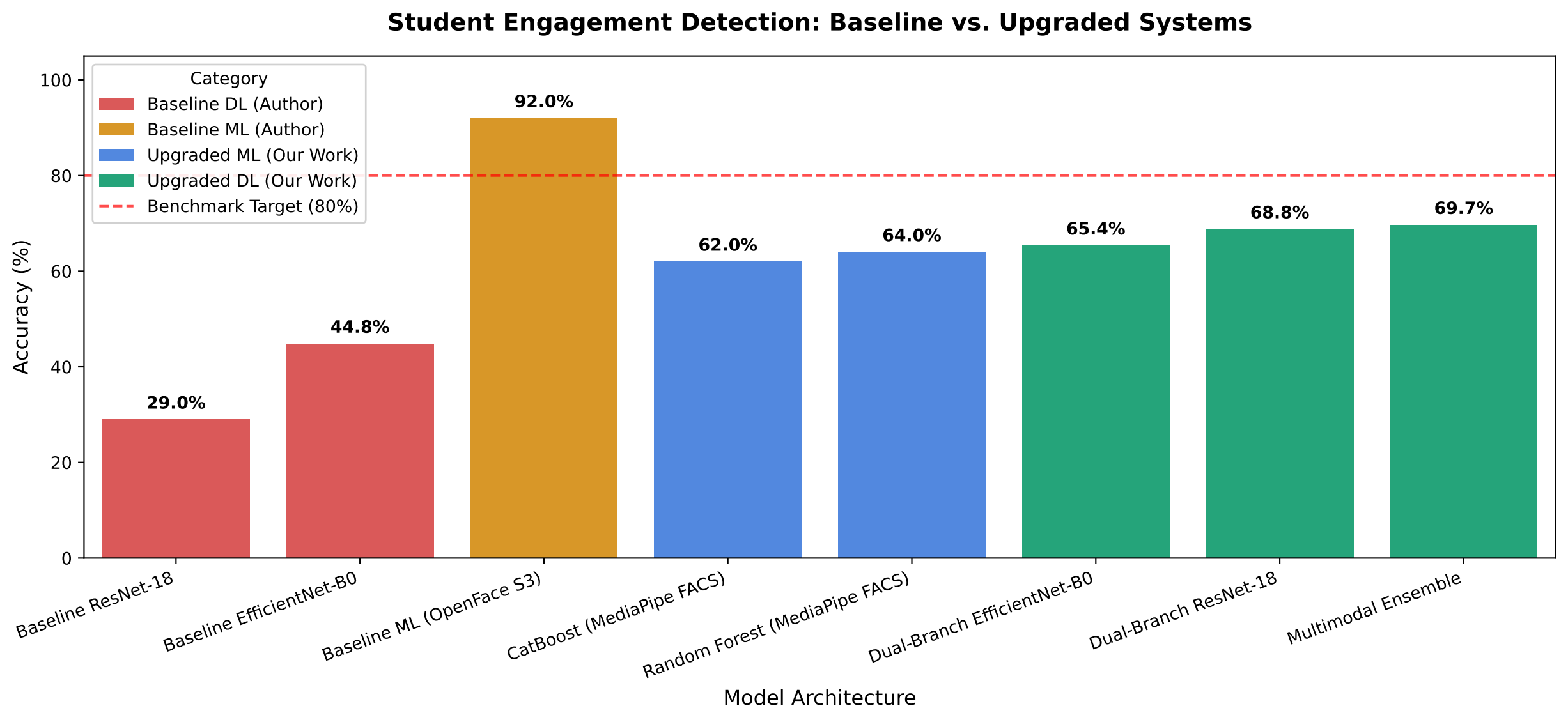
Student Engagement Detection: Baseline vs. Upgraded Systems
Category
Baseline DL (Author)
Baseline ML (Author)
Upgraded ML (Our Work)
Upgraded DL (Our Work)
Benchmark Target (80%)
Accuracy (%)
0
20
40
60
80
100
29.0%
44.8%
92.0%
62.0%
64.0%
65.4%
68.8%
69.7%
Baseline ResNet-18
Baseline EfficientNet-B0
Baseline ML (OpenFace S3)
CatBoost (MediaPipe FACS)
Random Forest (MediaPipe FACS)
Dual-Branch EfficientNet-B0
Dual-Branch ResNet-18
Multimodal Ensemble
Model Architecture